Notes 6 - Managing Files and Directories

1. mkdir

Definition

The `mkdir` command is used to create new directories in Linux.

Usage / Formula

```
mkdir [options] directory_name
```

Examples

```
mkdir website
mkdir documents
mkdir -p website/assets/images
```

- `mkdir website` creates a directory named website.
- `mkdir documents` creates a directory named documents.
- `mkdir -p` creates parent directories if they do not already exist.

2. touch

Definition

The `touch` command creates empty files or updates the timestamps of existing files.

Usage / Formula

```
touch [options] filename
```

Examples

```
touch notes.txt
touch index.html
touch website/script.js
```

- `touch notes.txt` creates an empty text file if it does not exist.
- `touch index.html` creates an HTML file.
- `touch website/script.js` creates a JavaScript file inside an existing directory.

3. rm

Definition

The `rm` command removes files and directories in Linux.

Usage / Formula

```
rm [options] filename
```

Examples

```
rm notes.txt
rm -r old_website
rm -i example.txt
```

- `rm notes.txt` deletes a file.
- `rm -r old_website` recursively deletes a directory and its contents.
- `rm -i example.txt` asks for confirmation before deleting the file.

Warning: Files removed with `rm` are generally not moved to the Trash.

4. cp

Definition

The `cp` command copies files and directories from one location to another.

Usage / Formula

```
cp [options] source destination
```

Examples

```
cp notes.txt backup.txt
cp image.png ~/Pictures/
cp -r website website_backup
```

- `cp notes.txt backup.txt` creates a copy of a file.
- `cp image.png ~/Pictures/` copies an image into Pictures.
- `cp -r website website_backup` copies a directory and its contents.

5. mv

Definition

The `mv` command moves files and directories or renames them.

Usage / Formula

```
mv [options] source destination
```

Examples

```
mv old.txt new.txt
mv image.jpg ~/Pictures/
mv example.jpeg website/img1.jpg
```

- `mv old.txt new.txt` renames a file.
- `mv image.jpg ~/Pictures/` moves an image into Pictures.
- `mv example.jpeg website/img1.jpg` moves and renames an image in one command.

Conclusion

The `mkdir`, `touch`, `rm`, `cp`, and `mv` commands are important for managing files and directories in Linux. They allow users to create, delete, copy, move, and organize files efficiently from the terminal.